

Chapter Six

Definite Descriptions

1 DEFINITE DESCRIPTIONS

A definite description is a singular phrase of English beginning with the definite article 'the', used as if to refer to a single thing. Examples are:

the winner of American Idol
the girl who won a medal
the book that Betty wrote
the assignment from last week
the even prime
the integer between 0 and 2

All theories about definite descriptions that treat them as terms agree on how to symbolize these phrases. First, there is a symbol called the *definite description operator*; it is the Greek letter iota (ι), rotated 180 degrees and often stylized as:

ι

This operator combines with a variable, just as a quantifier sign does, and the operator plus variable combines with a formula, as a quantifier does. But instead of making another formula, it makes a complex term. To symbolize the definite description '*the book that Betty wrote*' we would put ' ιx ' on the front of the formula ' x is a book and Betty wrote x ' to make:

$\iota x[Bx \wedge W(bx)]$

We can read ' ιx ' as '*the thing such that*', and so we can read that definite description as:

the thing such that it is a book and Betty wrote it

Other examples are:

<i>the winner of American Idol</i>	$\iota xW(xa)$	
<i>the girl who won a medal</i>	$\iota x[Gx \wedge \exists y[My \wedge W(xy)]]$	
<i>the even prime</i>	$\iota x[Ex \wedge \iota x]$	
<i>the integer between 0 and 2</i>	$\iota x[\iota x \wedge B(x02)]$	$B(\textcircled{1}\textcircled{2}\textcircled{3})$: ① is between ② and ③

There are different theories concerning the exact logical status of definite descriptions. We focus here on what is probably the simplest account, according to which definite descriptions are complex terms, on a par with proper names and with terms built up using operation symbols.

Formation rule:

Definite descriptions

If ' $\square$ ' is a formula, ' $\iota x\square$ ' is a term.

On this view, definite descriptions occur in formulas exactly where terms may occur. Some examples of formulas that contain definite descriptions and the definite descriptions that they contain are:

$A \iota x Fx$	$\iota x Fx$	
$S(\iota y Fy \iota z Gz)$	$\iota y Fy$	$\iota z Gz$
$B \iota x [Fx \wedge Gx] \rightarrow C \iota x Hx$	$\iota x [Fx \wedge Gx]$	$\iota x Hx$
$B \iota x \neg Dx \vee \neg B \iota x R(xa)$	$\iota x \neg Dx$	$\iota x R(xa)$
$B \iota x R(x \iota y S(xy))$	$\iota x S(xy)$	$\iota x R(x \iota y S(xy))$ <the latter contains the former>
$a = \iota x R(ax)$	$\iota x R(ax)$	
$\neg \iota x Fx = \iota x Gx$	$\iota x Fx$	$\iota x Gx$
$\forall x \forall y [A \iota z S(zx) \wedge B \iota u S(yu)]$	$\iota z S(zx)$	$\iota u S(yu)$

EXERCISES

1. Which of the following are well-formed formulas?

$$B \sim \neg xFx$$

$$\sim B \neg xFx$$

$$H(\neg xFx \neg xFx)$$

$$\forall x \forall y (x = \neg z \exists u (G(zu) \wedge Au) \wedge G(y \neg v Cv) \rightarrow x = y)$$

$$\neg z (Bz \wedge \exists y R(zy)) = \neg u Du$$

$$\neg \neg Ax = \neg \neg Ax$$

$$\neg x \neg \neg Ax = \neg \neg y By$$

$$R(\neg x R(x \neg y Fy)y)$$

2 SYMBOLIZING SENTENCES WITH DEFINITE DESCRIPTIONS

Symbolizing a sentence containing a definite description involves two tasks. One task is to figure out how to construct the definite description. The other task is figuring out where to put the definite description in the symbolization of the sentence containing it. This second task is easy; you just treat the definite description as you would any other term, simple or complex. So you put the definite description in the same place that you would put a name, if there were a name instead of the definite description. For example, if you were symbolizing '*Anna sees the giraffe*', you could consider instead how to symbolize '*Anna sees Fido*'. If you do this you would write:

$$S(af)$$

You get the symbolization of '*Anna sees the giraffe*' by putting the definite description for '*the giraffe*' in place of 'f':

$$S(a \neg x Gx)$$

In more complex cases the same principle applies. If you are wondering how to symbolize '*Every giraffe that Anna owns likes the ferocious hyena*' then just ask yourself how to symbolize a sentence with '*Fido*' in place of '*the ferocious hyena*':

Every giraffe that Anna owns likes Fido

$$\forall x (Gx \wedge O(ax) \rightarrow L(xf))$$

Then put ' $\neg x(Fx \wedge Hx)$ ' in place of 'f':

$$\forall x (Gx \wedge O(ax) \rightarrow L(x \neg y (Fy \wedge Hy)))$$

where the variable 'y' has been used in the definite description instead of 'x' to avoid confusion with the 'x' in the quantifier. (Actually, it would be OK to use 'x' in this case. It is advisable to use 'y' so you don't have to figure out whether there might be a problem.)

The other part of symbolizing a definite description is determining what its contents should be. This can be done if one considers what the part after the ' $\neg x$ ' would be if the sentence you are symbolizing had a quantifier word such as '*every*' instead of '*the*'. Suppose the sentence is:

Anna likes the ferocious giraffe

and you want to know how to symbolize the '*ferocious giraffe*' part of the description. In such a case consider how you would symbolize a sentence containing '*every ferocious giraffe*'. Your symbolization would contain a form like this:

$$\forall x (\underline{Fx \wedge Gx} \rightarrow \square)$$

The formula to use for '*ferocious giraffe*' in the definite description is what occurs in the underlined place

above. So you would write:

$$L(a\iota x(Fx \wedge Gx))$$

In symbolizing sentences it can be useful to have a way to read the symbolized version. Recall the routine given in chapter 3 section 4:

"We can read a quantified formula by using this recipe:

Read any universal quantifier as "everything is such that", while reading any variable that it binds as a pronoun which has that 'everything' as its antecedent.
Read any existential quantifier as "something is such that" while reading any variable that it binds as a pronoun which has the 'something' as its antecedent.

Some examples of this are:

$\forall x Gx$	<i>everything is such that it is a giraffe</i>
$\exists x (Gx \wedge Ex)$	<i>something is such that it is a giraffe and it will run for reelection</i>
$\forall x (Gx \rightarrow Ex)$	<i>everything is such that if it is a giraffe then it will run for reelection</i>

We can add a provision for definite descriptions to the recipe given in the box above:

Read any prefix of the form ' ιx ' as "the thing such that", while reading any variable that it binds as a pronoun which has 'thing' as its antecedent.

So this formula:

$$L(a\iota y(Fy \wedge Hy))$$

can be read as follows:

L	$(a$	ιy	$(Fy$	$\wedge$	$Hy))$
Anna likes	the thing such that	it is ferocious	and		it is a hyena

Some examples of sentences that can be symbolized with definite descriptions are:

The cat that Maria sees is larger than the dog that she sees.

$$L(\iota x[Cx \wedge S(mx)] \iota x[Dx \wedge S(mx)])$$

No dog chased the cat that scratched it.

$$\sim \exists x[Dx \wedge H(x \iota y[Cy \wedge S(yx)])]$$

Everyone parked in the space that they saw.

$$\forall x[Ex \rightarrow P(x \iota y[Ay \wedge S(xy)])]$$

E①: ① is a person

A①: ① is a space

P(①②): ① parked in ②

S(①②): ① saw ②

Definite descriptions are often used to symbolize possessive constructions, using 'have' to indicate possession. For example, 'Fred's car' means 'the car that Fred has'. The sentence 'Maria saw Fred's car' could be symbolized:

$$S(m \iota x H(fx))$$

Definite descriptions are also naturally used with superlative constructions. The strong reading of the phrase 'the tallest coat rack' means something like 'the coat rack which is taller than every other coat rack':

$$\iota x[Cx \wedge \forall y[Cy \wedge y \neq x \rightarrow T(xy)]]$$

Maria put her fez on the tallest coat rack.

$$P(m \iota x[Fx \wedge H(mx)] \iota x[Cx \wedge \forall y[Cy \wedge y \neq x \rightarrow T(xy)]])$$

P(①②③): ① put ② on ③

Everyone parked in the largest space they saw.

$\forall x[Ex \rightarrow P(x \neg y[Ay \wedge S(xy) \wedge \forall z[z \neq y \wedge Az \wedge S(xz) \rightarrow L(yz)])]]$

$E\textcircled{1}$: $\textcircled{1}$ is a person

$P(\textcircled{1}\textcircled{2})$: $\textcircled{1}$ parked in $\textcircled{2}$

$L(\textcircled{1}\textcircled{2})$: $\textcircled{1}$ is larger than $\textcircled{2}$

EXERCISES

1. Symbolize each of the following,

Anna dated the tallest spy.

The person who put a bug in my drink will pay.

Beatrice likes the man who bought her a ring.

Every giraffe loves the keeper who feeds it.

Every giraffe loves the tallest keeper who feeds it.

The woman who studied did better than the woman who didn't study.

Everybody honors the woman who gave birth to her/him.

The prize will be awarded to the person who spells the word correctly.

Every woman parked her own car.

2. Read each symbolized sentence in stilted English using the recipe given above.

3 DERIVATIONAL RULES FOR DEFINITE DESCRIPTIONS: PROPER DESCRIPTIONS

Some definite descriptions are "proper", and some "improper". A proper definite description is one whose descriptive part correctly describes exactly one thing. A definite description is improper when its descriptive part either describes more than one thing, or describes nothing at all. For example, the definite description 'the book that Betty wrote', is proper if Betty wrote exactly one book; if she wrote several books, or didn't write any at all, it is improper. In symbols, we can say about any definite description:

' $\iota x \phi$ ' is proper if and only if this is true: $\exists z \forall x [\phi \leftrightarrow x=z]$
 $\xrightarrow{\text{same formula}}$

[For this relationship to hold, the variable 'z' must not occur in the formula ' ϕ '.]

For example, for the definite description that symbolizes 'the book that Betty wrote':

$\iota x [Bx \wedge W(bx)]$

we have:

' $\iota x [Bx \wedge W(bx)]$ ' is proper if and only if this is true: $\exists z \forall x [Bx \wedge W(bx) \leftrightarrow x=z]$
 $\xrightarrow{\text{same formula}}$

All theories that treat definite descriptions as terms agree on how to treat proper definite descriptions. They obey the rule that if the description is proper, the descriptive part is true when its variables (the occurrences bound by ' ι ') are replaced by the definite description itself. For example, if 'the book that Betty wrote' is proper, then the book that Betty wrote is indeed a book, and Betty did write it:

$\exists z \forall x [Bx \wedge W(bx) \leftrightarrow x=z]$ properness of 'book that Betty wrote'
 $\therefore B \iota x [Bx \wedge W(bx)] \wedge W(b \iota x [Bx \wedge W(bx)])$ the book that Betty wrote is a book $\wedge$ Betty wrote the book that Betty wrote

The pattern is always the same: if there is a unique such-and-such, then *the such-and-such* is such-and-such. When 'such-and-such' is complex, so is the application of this rule.

Rule for Proper Descriptions (prd)

If there is an available line or premise stating that the definite description ' $\iota x \phi$ ' is proper:

$\exists z \forall x [\phi \leftrightarrow x=z]$

then one may infer the formula that you get by taking ' ϕ ' alone, replacing every free occurrence of 'x' in it by ' $\iota x \phi$ '. As justification cite the earlier line number plus 'prd'.

<Constraints: This rule is subject to the restriction that 'z' is not free in ' ϕ ', and that no variable that is free in ' $\iota x \phi$ ' gets bound when ' $\iota x \phi$ ' is put in place of a free occurrence of 'x' in ' ϕ '.>

For example, above we saw that the definite description

$\iota x [Bx \wedge W(bx)]$

is proper, because this is true:

$\exists z \forall x [Bx \wedge W(bx) \leftrightarrow x=z].$

Applying the rule, we infer the result of replacing every free occurrence of 'x' in ' $Bx \wedge W(bx)$ ' by ' $\iota x [Bx \wedge W(bx)]$ '; that is, we infer:

$B \iota x [Bx \wedge W(bx)] \wedge W(b \iota x [Bx \wedge W(bx)])$.

EXERCISES

1. What can be inferred from the statements that say that these definite descriptions are proper?

The spy who loved me

The tallest giraffe to fly to the moon

The number whose square root is the same as its cube root

The boy such that he and the girl who saw him both sang

The largest gift given to UCLA

The big blue tuba

4 SYMBOLIZING ORDINARY LANGUAGE

Often when we use a definite description in speech the description is proper only when limited to what is under discussion when it is used. For example, if Maria owns a dog, we may, in speaking to her, say 'The dog is hungry'. Here we do not intend to be speaking about all animals on earth, or even all animals in town. If we were, the description, 'the dog', namely, ' $\lambda x D x$ ', would not be proper -- for there exist many dogs, not just one. However, if it is clear in the context in which we say 'the dog' that we are only speaking about things of immediate concern to us, then among those things there may indeed be exactly one thing that is a dog, and this would make our definite description proper.

When Frege introduced notation for definite descriptions it was within a project of providing a logical foundation for mathematics. Typically, before introducing a definite description he would prove it to be proper. This approach worked well for his enterprise. When it comes to using logical notation to symbolize statements made in ordinary language, things are different. For example, suppose that somebody says:

The dog that Betty owns chased the cat that Fred owns.

One would usually infer from such an utterance that Betty owns a dog, that Fred owns a cat, that a dog chased a cat, and many other things. This is because using definite descriptions in such a sentence usually presupposes that the definite descriptions are proper. But if you symbolize the sentence above in the most straightforward way in our current logical notation, there is no such assumption. That is, if you symbolize the sentence above as:

$H(\text{the dog that Betty owns, the cat that Fred owns})$

i.e. $H(\lambda x [D x \wedge O(bx)] \lambda x [C x \wedge O(fx)])$

there is nothing in the symbolization to indicate that the definite descriptions are proper. As a result, you cannot infer, for example, that some dog chased some cat:

$\exists x \exists y [D x \wedge C y \wedge H(xy)]$

So if you want to symbolize everything that is communicated by a use of the English sentence above, you will need to add to your symbolization the assumption that the description is proper, something like:

$H(\lambda x [D x \wedge O(bx)] \lambda x [C x \wedge O(fx)]) \wedge \text{'}\lambda x [D x \wedge O(bx)]\text{' is proper} \wedge \text{'}\lambda x [C x \wedge O(fx)]\text{' is proper}$

which in the case under discussion will be:

$H(\lambda x [D x \wedge O(bx)] \lambda x [C x \wedge O(fx)]) \wedge \exists z \forall x [D x \wedge O(bx) \leftrightarrow x=z] \wedge \exists z \forall x [C x \wedge O(fx) \leftrightarrow x=z]$

On the other hand, not every use of a definite description presupposes that the description is proper. Consider the negation of the above statement:

The dog that Betty owns didn't chase the cat that Fred owns.

One could reasonably follow up this assertion with "*in fact, Betty doesn't even own a dog!*". Some people think the displayed sentence can be used either so as to presuppose that the descriptions are proper, or used so as not to presuppose this. There is no real consensus on this issue.

EXERCISES

Symbolize each of the following arguments (i) so as to include the claim that the definite descriptions are proper, and then (ii) so as not to include that claim. Assess each symbolized argument; if it is valid, produce a derivation to show this; if not, produce a counterexample. [You may want to read section 6 below before producing the counter-example.]

1. *The hyena that a lion chased fled.*
 $\therefore$ *A hyena fled.*
2. *The winner will congratulate the loser.*
 $\therefore$ *There will be a winner.*
3. *The cat that Maria owns chased a mouse that ate the fig.*
 $\therefore$ *A cat that Maria owns chased a mouse that ate a fig.*

5 DERIVATIONAL RULES FOR DEFINITE DESCRIPTIONS: IMPROPER DESCRIPTIONS

Any complete theory of definite descriptions has to say how to handle improper definite descriptions -- descriptions whose descriptive part is not satisfied by anything at all, such as:

the planet between Mercury and Venus

or descriptions that are satisfied by more than one thing, such as:

the planet between Earth and the sun.

One natural response is that these are terms that don't refer to anything at all. Some systems of logic work in this way -- they take improper definite descriptions to be terms that fail to refer. This is certainly a feasible approach, but it is a complicated one if we want a way to treat improper definite descriptions that fits in with the rules that we already have in this text. From chapter 3 on, we have made the idealization that our closed terms (simple names and complex terms containing operation symbols) each refer to a single thing. We will continue that idealization here, in the understanding that in the case of definite descriptions this is clearly artificial.

Our artificial technique for handling improper definite descriptions was suggested over a century ago by the logician Gottlob Frege. The technique is to arbitrarily choose something for all improper descriptions to stand for. We then assume that any improper definite description stands for this thing. This thing can be anything -- the number zero, your pet dog, the tallest giraffe in the San Diego Zoo, the left front burner of the stove on which I cooked oatmeal today. Since the thing is arbitrarily chosen, we will not identify it in any further way, say by assigning a simple name to it, or applying a predicate to it. Any name that we are using might actually name the artificially chosen thing, but nothing in our logic tells us so.

In spite of not knowing what it is, we can easily refer to this arbitrarily chosen thing with an appropriate complex term. We just need to use a definite description that we know to be improper. A natural example is to use the definite description:

$\neg \exists x x \neq x$ *the thing that is not identical to itself*

Since nothing can fail to be identical to itself, this definite description has to be improper. This is a logical truth, for the statement that the description is improper is:

$$\therefore \neg \exists z \forall x [x \neq x \leftrightarrow x = z]$$

and we can easily produce a derivation to show that this is a theorem of logic:

1.	Show $\neg \exists z \forall x [x \neq x \leftrightarrow x = z]$	
2.	$\exists z \forall x [x \neq x \leftrightarrow x = z]$	ass id
3.	$\neg x x \neq x \neq \neg x x \neq x$	2 prd
4.	$\neg x x \neq x = \neg x x \neq x$	sid
5.		3 4 id

← putting ' $\neg x x \neq x$ ' in for both occurrences of 'x' in ' $x \neq x$ '.

Our single rule for improper definite descriptions says that if a definite description is improper, it refers to whatever ' $\neg x x \neq x$ ' refers to:

Rule for Improper Descriptions (imd)

If there is a statement on an available line or premise stating that the definite description ' $\neg x \circ$ ' is improper:

$$\neg \exists z \forall x [\circ \leftrightarrow x = z]$$

Then one may infer:

$$\neg x \circ = \neg x x \neq x$$

Justification: cite the earlier line number plus 'imd'

<As before, the variable 'z' must not be free in ' $\circ$ '.>

A description must either be proper or improper, although based on information given to us we may not know which. We do know this much, however: if the definite description does *not* refer to the chosen object, it must be proper:

$$\neg x \circ \neq \neg x x \neq x \quad \therefore \exists z \forall x [x \leftrightarrow x = z]$$

This is a trivial consequence of the rule for improper descriptions:

1.	Show $\exists z \forall x [x \leftrightarrow x = z]$	
2.	$\neg \exists z \forall x [x \leftrightarrow x = z]$	ass id
3.	$\neg x \circ = \neg x x \neq x$	2 imd
4.	$\neg x \circ \neq \neg x x \neq x$	pr1
5.		3 4 id

One must be careful not to make a similar but invalid inference. Given that

$$\neg x \circ = \neg x x \neq x$$

one may not infer from this that ' $\neg x \circ$ ' is improper. Since the chosen object may be anything at all, it might be the dog that Cynthia bought. In this case the definite description 'the dog that Cynthia bought' refers properly to the chosen object. That is, we have:

$$\neg x [Dx \wedge B(cx)] = \neg x x \neq x$$

where the definite description ' $\neg x [Dx \wedge B(cx)]$ ' is proper:

$$\exists z \forall x [Dx \wedge B(cx) \leftrightarrow x = z]$$

So the chosen object can be referred to by proper descriptions, in addition to improper ones.

Some applications of the rule for improper descriptions are relatively straightforward. An example is:

$$\exists x \exists y [x \neq y \wedge Fx \wedge Fy]$$

$$\neg \exists x Gx$$

$$\therefore \neg x Fx = \neg x Gx$$

1.	Show $\neg x Fx = \neg x Gx$	
2.	Show $\neg \exists z \forall x [Fx \leftrightarrow x = z]$	
3.	$\exists z \forall x [Fx \leftrightarrow x = z]$	ass id
4.	$\forall x [Fx \leftrightarrow x = i]$	3 ei
5.	$u \neq v \wedge Fu \wedge Fv$	pr1 ei ei
6.	$Fu \leftrightarrow u = i$	4 ui
7.	$u = i$	5 s s 6 bp
8.	$Fv \leftrightarrow v = i$	4 ui
9.	$v = i$	5 s 8 bp
10.	$u = v$	7 9 LL
11.	$u \neq v$	5 s s 10 id
12.	Show $\neg \exists z \forall x [Gx \leftrightarrow x = z]$	
13.	$\exists z \forall x [Gx \leftrightarrow x = z]$	ass id
14.	$\forall x [Gx \leftrightarrow x = j]$	13 ei
15.	$Gj \leftrightarrow j = j$	14 ui
16.	Gj	sid 15 bp
17.	$\exists x Gx$	16 eg
18.	$\neg \exists x Gx$	pr2 17 id
19.	$\neg x Fx = \neg x x \neq x$	2 imd
20.	$\neg x Gx = \neg x x \neq x$	12 imd
21.	$\neg x Fx = \neg x Gx$	19 20 LL dd

← rule imd

← rule imd

Often you will be given an argument whose premises contain definite descriptions that may be either proper or improper. If you can prove that a definite description is proper, you can often use that to prove other desired things. Likewise, if you can prove that a definite description is improper, you can often use that to prove other desired things. But sometimes you cannot prove either of these things, because not enough information is given to decide. You may still be able to use *both* strategies just described: you must both (i) infer what you want to infer using the assumption that the definite description is proper, and also (ii) infer what you want to infer using the assumption that the definite description is improper. If you can do this, you can use the rule for separation of cases to get the desired conclusion.

	$\forall x[Hx \rightarrow Gx]$	
	$F1x \rightarrow G1x$	
	$\sim H1x \rightarrow 1x \neq 1x$	
	$\therefore \exists xGx$	
1.	Show $\exists xGx$	
2.	Show $\exists z \forall x[Fx \leftrightarrow x=z] \rightarrow \exists xGx$	
3.	$\exists z \forall x[Fx \leftrightarrow x=z]$	ass cd
4.	$F1x$	3 prd
5.	$G1x$	pr2 4 mp
6.	$\exists xGx$	5 eg cd
7.	Show $\sim \exists z \forall x[Fx \leftrightarrow x=z] \rightarrow \exists xGx$	
8.	$\sim \exists z \forall x[Fx \leftrightarrow x=z]$	ass cd
9.	$1x \neq 1x$	8 imd
10.	$H1x$	9 dn pr3 mt
11.	$G1x$	pr1 ui 10 mp
12.	$\exists xGx$	11 eg cd
13.	$\exists xGx$	2 7 sc dd ← rule sc

[You may be tempted to use separation of cases with the cases being identity with the chosen object (" $1x \neq 1x$ ") and non-identity with the chosen object (" $1x \neq 1x$ "). But this is not often useful, because if a definite description is identical to the chosen object, you still don't know whether it is proper or not, and so you can't use either rule prd or rule imd. It is usually better to take the cases to be the statements that the definite description is proper, and that it is improper.]

EXERCISES

Produce derivations for the following arguments.

- $\exists x[Fx \wedge Gx \wedge Hx]$
 $\exists x[Fx \wedge Gx \wedge \sim Hx]$
 $\therefore 1x \neq 1x$
- $Fa \wedge Fb \wedge a \neq b$
 $\forall x[Fx \wedge x \neq a \leftrightarrow Gx]$
 $1x \neq 1x$
 $\therefore \forall x[Gx \rightarrow x = b]$
- $\forall x \exists y[R(xy) \wedge \forall z[z \neq y \rightarrow \sim R(xz)]]$
 $\exists x \exists y[x \neq y \wedge \forall z[z = x \vee z = y]]$
 $\therefore 1x \neq 1x$

6 INVALIDITIES WITH DEFINITE DESCRIPTIONS

The technique of giving counter-examples to show invalidity is unchanged from previous chapters, with one exception. Since what ' $\iota x x \neq x$ ' stands for is not determined by its wording, when a counter-example is given the reference of ' $\iota x x \neq x$ ' must be independently specified. Since rearranging things in the universe has no effect on counter-examples, it is simplest to always choose 0 to be the chosen object. We will abide by that custom here.

Here are some counter-examples for arguments containing definite descriptions. The reader should review each to see how the counter-example works.

$$\begin{aligned} & \exists z \forall x [Fx \wedge Gx \leftrightarrow x=z] \\ & \exists x [Fx \wedge \neg Gx] \\ & \therefore \exists y [Gy \wedge y = \iota x [Fx \wedge Gx]] \end{aligned}$$

Universe: {0, 1, 2}
F: {1, 2}
G: {1}
 $\iota x x \neq x$: 0

The first premise is true because the conjunction ' $Fx \wedge Gx$ ' is true of exactly one thing. The second premise is true because there is something, namely 2, that is F but not G. The conclusion is false because there isn't anything that is G and unequal to 1, which is what the proper description, ' $\iota x [Fx \wedge Gx]$ ', stands for.

$$\begin{aligned} & \forall y \exists z \forall x [R(xy) \leftrightarrow x=z] \\ & \therefore \exists y y = \iota x R(xy) \end{aligned}$$

Universe {0, 1}
R: {<0, 1>, <1, 0>}
 $\iota x x \neq x$: 0

The premise is true because whatever you pick for y, there is something -- namely, whatever y isn't -- such that everything bears R to y iff it is that thing that y isn't. The conclusion is false because nothing is the unique thing that is related to it by R: 0 isn't such a thing, and 1 isn't such a thing.

$$\begin{aligned} & \neg \forall x [Fx \leftrightarrow Gx] \\ & \therefore \iota x Fx \neq \iota x Gx \end{aligned}$$

Universe: {0, 1, 2}
F: {}
G: {1, 2}
 $\iota x x \neq x$: 0

The premise is true because F and G don't agree everywhere. (In fact, they agree nowhere.) The conclusion is false because both Fx and Gx are improper -- the first is true of nothing, and the second is true of two things -- so both definite descriptions stand for the chosen object, 0.

EXERCISES

Give counter-examples to show that these arguments are invalid:

1. $\forall x Ax$
 $\therefore \forall x \forall y (Ax \wedge Ay \rightarrow x=y)$
2. $\forall x \exists y R(xy)$
 $\exists x \forall y [R(xy) \rightarrow y=a]$
 $\therefore \iota x R(xx) = \iota x x \neq x$
3. $\iota x Ax \neq \iota x Bx$
 $\iota x Bx \neq \iota x Cx$
 $d(a) = \iota x Bx$
 $\forall x d(d(x)) = d(x)$
 $\therefore \iota x Ax \neq \iota x Cx$

7 COUNTER-EXAMPLES WITH INFINITE UNIVERSES

There are invalid arguments containing definite descriptions whose counter-examples require infinite universes. No new techniques are involved in giving such counter-examples.

Here is an example of an invalid argument that can't be given a counter-example with a finite universe:

$$\begin{array}{l} \forall y \exists z \forall x [R(yx) \leftrightarrow x=z] \\ \forall x \forall y [x \neq y \rightarrow \exists z R(xz) \neq \exists z R(yz)] \\ \therefore \forall x \exists y R(yx) \end{array}$$

A counter-example that shows it to be invalid is:

$$\begin{array}{l} \text{Universe: } \{0, 1, 2, \dots\} \\ R(\textcircled{1}\textcircled{2}): \textcircled{2} = \textcircled{1} + 1 \end{array}$$

The first premise is true here because for whatever you pick for y , there is something -- namely, y 's successor -- such that everything is one greater than y iff it is y 's successor. The second premise is true because whenever there are two different things, the unique thing that is one greater than the first is something other than the unique thing that is one greater than the second. And the conclusion is false because 0 is such that there is nothing such that it plus 1 is 0.

EXERCISES

Give counter-examples with infinite universes to show that these arguments are invalid:

1. $\forall x \forall y [x \neq y \rightarrow d(x) \neq d(y)]$
 $\therefore \neg x \sim \exists y x = d(y) = \neg x x \neq x$
2. $\forall x R(xc(x))$
 $\forall x \forall y \forall z [R(xy) \wedge R(yz) \rightarrow R(xz)]$
 $\therefore \forall x \exists y x = \exists z R(yz)$
3. $\forall x \forall y [b(x) = b(y) \rightarrow x = y]$
 $\exists y \forall x \sim y = \exists z [z = b(x)]$
 $\therefore \exists x \exists y [x \neq y \wedge \forall z [b(z) \neq x \wedge b(z) \neq y]]$

Chapter Six -- Answers to the exercises

1 DEFINITE DESCRIPTIONS

1. Which of the following are well-formed formulas?

$B \sim \iota x Fx$	no	negation cannot directly precede a term
$\sim B \iota x Fx$	yes	
$H(\iota x Fx \iota x Fx)$	yes	
$\forall x \forall y (x = \iota z \exists u (G(zu) \wedge Au) \wedge G(y \iota v Cv) \rightarrow x = y)$	yes	
$\iota z (Bz \wedge \exists y R(zy)) = \iota u Du$	yes	
$\iota \sim Ax = \sim \iota Ax$	no	no variable following first ι
$\iota x \sim \sim Ax = \sim \iota y By$	no	negation cannot directly precede a term
$R(\iota x R(x \iota y Fy)y)$	yes	

2 SYMBOLIZING SENTENCES WITH DEFINITE DESCRIPTIONS

1. Symbolize each of the following,

2. Read each symbolized sentence in stilted English using the recipe given above.

Anna dated the tallest spy.

$D(a \iota x [Fx \wedge \forall y [Fy \wedge y \neq x \rightarrow T(xy)]]])$

F: spy

Anna dated the thing such that it is a spy and everything is such that if it is a spy and different from it then it is taller than it

The person who put a bug in my drink will pay.

$\iota x [Ex \wedge \exists y [By \wedge P(xyz) [Dz \wedge H(az)]]]$ E: person P(xyz): x put y in z a: me H: have I: will pay

The thing such that it is a person and something is such that it is a bug and it put it in the thing such that it is a drink and I have it will pay

Beatrice likes the man who bought her a ring.

$L(b \iota x [Mx \wedge \exists y [\iota y \wedge B(xby)]]])$

Beatrice likes the thing such that it is a man and something is such that it is a ring and he bought her it

Every giraffe loves the keeper who feeds it.

$\forall x [Gx \rightarrow L(x \iota y [Ky \wedge F(yx)])]$

Everything is such that if it is a giraffe then it likes the thing such that it is a keeper and it feeds it

Every giraffe loves the tallest keeper who feeds it.

$\forall x [Gx \rightarrow L(x \iota y [Ky \wedge F(yx) \wedge \forall z [z \neq y \wedge Kz \wedge F(zx) \rightarrow T(yz)])]]]$

Everything is such that if it is a giraffe then it likes the thing such that it is a keeper and feeds it and everything is such that if it is another keeper who feeds it then it is taller than it

The woman who studied did better than the woman who didn't study.

$B(\iota x [Ax \wedge Ex] \iota x [Ax \wedge \sim Ex])$ B: did better than A: woman E: studied

The thing such that it is a woman and it studied did better than the thing such that it is a woman and it didn't study.

Everybody honors the woman who gave birth to her/him.

$\forall x [Ex \rightarrow H(x \iota y [Ay \wedge B(yx)])]$

Everything is such that if it is a person then it honors the thing such that it is a woman and it gave birth to it.

The prize will be awarded to the person who spells the word correctly.

$A(\iota x \iota y [\iota y \wedge S(y \iota z Fx)])$ I: prize A: will be awarded to S: spells correctly F: word

The thing such that it is a prize will be awarded to the thing such that it is a person and it spells correctly the thing such that it is a word.

Every woman parked her own car.

$\forall x [Ax \rightarrow P(x \iota y [Cy \wedge H(xy)])]$

Everything is such that if it is a woman then it parked the thing such that it is a car and she has it.

3 DERIVATIONAL RULES FOR DEFINITE DESCRIPTIONS: PROPER DESCRIPTIONS

1. What can be inferred from the statements that say that these definite descriptions are proper?

The spy who loved me

There is one and only one spy who loved me.

The tallest giraffe to fly to the moon

There is one and only one giraffe which flew to the moon and is taller than every other giraffe which flew to the moon.

The number whose square root is the same as its cube root

There is one and only one number such that its square root is its cube root.

<Note that this is false, since the condition is met by both 0 and 1>

The boy such that he and the girl who saw him both sang

There is one and only one thing such that it is a boy and there is one and only one girl who saw it and they both sang.

The largest gift given to UCLA

There is one and only one gift given to UCLA which is larger than any other gift given to UCLA.

The big blue tuba

There is one and only one thing which is big and blue and is a tuba.

4 SYMBOLIZING ORDINARY LANGUAGE

Symbolize each of the following arguments (i) so as to include the claim that the definite descriptions are proper, and then (ii) so as not to include that claim. Assess each symbolized argument; if it is valid, produce a derivation to show this; if not, produce a counterexample. [You may want to read section 6 below before producing the counter-example.]

1. *The hyena that a lion chased fled.*

∴ *A hyena fled.*

(i) $\neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]]$

$\exists z \forall x [Hx \wedge \exists y [ly \wedge C(yx)] \leftrightarrow x=z]$ <properness of ' $\neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]]$ '>

∴ $\exists x [Hx \wedge Fx]$

1. **Show** $\exists x [Hx \wedge Fx]$

2.	$H \neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]] \wedge \exists y [ly \wedge C(y \neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]])]$	pr2 prd
3.	$H \neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]] \wedge \neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]]$	2 s pr1 adj
4.	$\exists x [Hx \wedge Fx]$	3 eg dd

(ii) $\neg \exists x [Hx \wedge \exists y [ly \wedge C(yx)]]$

∴ $\exists x [Hx \wedge Fx]$

INVALID

Universe: {0,1}

F: {0}

H: {}

I: {}

C: {}

chosen object: 0

The premise is true because the description is improper, so it refers to 0, which 'F' is true of.

The conclusion is false because nothing is both H and F.

2. *The winner will congratulate the loser.*

$\therefore$ *There will be a winner.*

(i) $C(\lambda x \lambda y \lambda z Ex)$

$\exists z \forall x [Ix \leftrightarrow x = z]$ <properness of ' $\lambda x Ix$ '>

$\exists z \forall x [Ex \leftrightarrow x = z]$ <properness of ' $\lambda x Ex$ '>

$\therefore \exists x Ix$

1. ~~Show~~ $\exists x Ix$

2.	$IxIx$	pr2 prd
3.	$\exists x Ix$	2 eg dd

(ii) $C(\lambda x \lambda y \lambda z Ex)$

$\therefore \exists x Ix$

INVALID

Universe: $\{0,1\}$

I: $\{\}$

E: $\{\}$

C: $\{<0,0>\}$

chosen object: 0

3. *The cat that Maria owns chased a mouse that ate the fig.*

$\therefore$ *A cat that Maria owns chased a mouse that ate a fig.*

(i) $\exists x [Ex \wedge A(x \lambda y Fy) \wedge H(\lambda u [Cu \wedge O(au)]x)]$ E: mouse A: ate H: chased a: Maria

$\exists z \forall x [Fx \leftrightarrow x = z]$ <properness of ' $\lambda y Fy$ '>

$\exists z \forall x [Cx \wedge O(ax) \leftrightarrow x = z]$ <properness of ' $\lambda u [Cu \wedge O(au)]$ '>

$\therefore \exists v [Cv \wedge O(av) \wedge \exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(vw)]]$

1. ~~Show~~ $\exists v [Cv \wedge O(av) \wedge \exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(vw)]]$

2.	$C\lambda u [Cu \wedge O(au)] \wedge O(a\lambda u [Cu \wedge O(au)])$	pr3 prd
3.	$Ei \wedge A(i\lambda y Fy) \wedge H(\lambda u [Cu \wedge O(au)]i)$	pr1 ei
4.	$F\lambda y Fy$	pr2 prd
5.	$A(i\lambda y Fy)$	3 s s
6.	$F\lambda y Fy \wedge A(i\lambda y Fy)$	4 5 adj
7.	$\exists r [Fr \wedge A(ir)]$	6 eg
8.	$Ei \wedge \exists r [Fr \wedge A(ir)]$	3 s s 7 adj
9.	$Ei \wedge \exists r [Fr \wedge A(ir)] \wedge H(\lambda u [Cu \wedge O(au)]i)$	3 s 8 adj
10.	$\exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(\lambda u [Cu \wedge O(au)]w)]$	9 eg
11.	$C\lambda u [Cu \wedge O(au)] \wedge O(a\lambda u [Cu \wedge O(au)]) \wedge$ $\exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(\lambda u [Cu \wedge O(au)]w)]$	2 10 adj
12.	$\exists v [Cv \wedge O(av) \wedge \exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(vw)]]$	11 eg dd

(ii) $\exists x [Ex \wedge A(x \lambda y Fy) \wedge H(\lambda u [Cu \wedge O(au)]x)]$

$\therefore \exists v [Cv \wedge O(av) \wedge \exists w [Ew \wedge \exists r [Fr \wedge A(wr)] \wedge H(vw)]]$

INVALID

Universe: $\{0, 1, 2\}$

F: $\{1\}$

E: $\{2\}$

A: $\{<2,1>\}$

C: $\{\}$

O: $\{\}$

H: $\{<0,2>\}$

chosen object: 0

5 DERIVATIONAL RULES FOR DEFINITE DESCRIPTIONS: IMPROPER DESCRIPTIONS

Produce derivations for the following arguments.

1. $\exists x[Fx \wedge Gx \wedge Hx]$
 $\exists x[Fx \wedge Gx \wedge \sim Hx]$
 $\therefore \neg \exists x Fx = \neg \exists x Gx$

1. **Show** $\neg \exists x Fx = \neg \exists x Gx$

2.	$Fi \wedge Gi \wedge Hi$	pr1 ei
3.	$Fj \wedge Gj \wedge \sim Hj$	pr2 ei
4.	Show $\sim \exists z \forall x [Fx \leftrightarrow x=z]$	
5.	$\exists z \forall x [Fx \leftrightarrow x=z]$	ass id
6.	$\forall x [Fx \leftrightarrow x=u]$	5 ei
7.	$Fi \leftrightarrow i=u$	6 ui
8.	$i=u$	2 s s 7 bp
9.	$Fj \leftrightarrow j=u$	6 ui
10.	$j=u$	3 s s 9 bp
11.	Hu	2 s 10 LL
12.	$\sim Hu$	3 s 10 LL 11 id
13.	Show $\sim \exists z \forall x [Gx \leftrightarrow x=z]$	
14.	$\exists z \forall x [Gx \leftrightarrow x=z]$	ass id
15.	$\forall x [Gx \leftrightarrow x=v]$	14 ei
16.	$Fi \leftrightarrow i=v$	15 ui
17.	$i=v$	2 s s 16 bp
18.	$Fj \leftrightarrow j=v$	6 ui
19.	$j=v$	3 s s 18 bp
20.	Hv	2 s 19 LL
21.	$\sim Hv$	3 s 19 LL 20 id
22.	$\neg \exists x Fx = \neg \exists x \neq x$	4 imd
23.	$\neg \exists x Gx = \neg \exists x \neq x$	13 imd
24.	$\neg \exists x Fx = \neg \exists x Gx$	22 23 LL dd

2. $Fa \wedge Fb \wedge a \neq b$
 $\forall x[Fx \wedge x \neq a \leftrightarrow Gx]$
 $\neg \exists x Gx \neq \neg \exists x Fx$
 $\therefore \forall x[Gx \rightarrow x=b]$

1. **Show** $\forall x[Gx \rightarrow x=b]$

2. **Show** $Gx \rightarrow x=b$

3.	Gx	ass cd
4.	Show $\neg \exists z \forall x[Fx \leftrightarrow x=z]$	
5.	$\exists z \forall x[Fx \leftrightarrow x=z]$	as id
6.	$\forall x[Fx \leftrightarrow x=i]$	5 ei
7.	$Fa \leftrightarrow a=i$	6 ui
8.	$a=i$	pr1 s s 7 bp
9.	$Fb \leftrightarrow b=i$	6 ui
10.	$b=i$	pr1 s s 9 bp
11.	$a=b$	8 10 LL
12.	$a \neq b$	pr1 s 11 id
13.	$\neg \exists x Fx = \neg \exists x x \neq x$	4 imd
14.	$\neg \exists x Gx \neq \neg \exists x x \neq x$	13 pr3 LL
15.	Show $\exists z \forall x[Gx \leftrightarrow x=z]$	
16.	$\neg \exists z \forall x[Gx \leftrightarrow x=z]$	ass id
17.	$\neg \exists x Gx = \neg \exists x x \neq x$	16 imd
18.	$\neg \exists x Gx \neq \neg \exists x x \neq x$	14 r 17 id
19.	$\forall x[Gx \leftrightarrow x=j]$	15 ei
20.	$Fb \wedge b \neq a \leftrightarrow Gb$	pr2 ui
21.	$b \neq a$	pr1 s sm
22.	$Fb \wedge b \neq a$	pr1 s s 21 adj
23.	Gb	20 22 bp
24.	$Gb \leftrightarrow b=j$	19 ui
25.	$b=j$	23 24 bp
26.	$Gx \leftrightarrow x=j$	19 ui
27.	$x=j$	3 26 bp
28.	$x=b$	25 27 LL cd
29.		2 ud

3. $\forall x \exists y [R(xy) \wedge \forall z [z \neq y \rightarrow \sim R(xz)]]$
 $\exists x \exists y [x \neq y \wedge \forall z [z = x \vee z = y]]$
 $\therefore \neg \exists x R(bx) \neq \neg \exists x \sim R(bx)$

1. Show $\neg \exists x R(bx) \neq \neg \exists x \sim R(bx)$

2. $i \neq j \wedge \forall z [z = i \vee z = j]$ pr2 ei ei

3. $R(bk) \wedge \forall z [z \neq k \rightarrow \sim R(bz)]$ pr1 ui ei

4. Show $\exists z \forall x [R(bx) \leftrightarrow x = z]$

show 'R(bx)' is proper

5. Show $\forall x [R(bx) \leftrightarrow x = k]$

6. Show $R(bx) \leftrightarrow x = k$

7. Show $R(bx) \rightarrow x = k$

$R(bx)$	ass cd
$x \neq k \rightarrow \sim R(bx)$	3 s ui
$\sim x \neq k$	8 dn 9 mt
$x = k$	10 dn cd

12. Show $x = k \rightarrow R(bx)$

$x = k$	ass cd
$R(bx)$	3 s 13 LL cd
$R(bx) \leftrightarrow x = k$	7 12 cb dd

16. $\exists z \forall x [R(bx) \leftrightarrow x = z]$ 6 ud

17. $\exists z \forall x [R(bx) \leftrightarrow x = z]$ 5 eg dd

18. Show $k = i \rightarrow \exists z \forall x [\sim R(bx) \leftrightarrow x = z]$

show that if $k = i$ then

' $\sim R(bx)$ ' is proper

19. $k = i$ ass cd

20. Show $\forall x [\sim R(bx) \leftrightarrow x = j]$

21. Show $\sim R(bx) \leftrightarrow x = j$ 2 s 19 LL

22. Show $\sim R(bx) \rightarrow x = j$

$\sim R(bx)$	ass cd
$x \neq k$	3 s 23 LL
$x \neq i$	19 24 LL
$x = j$	2 s ui 25 bp cd

27. Show $x = j \rightarrow \sim R(bx)$

$x = j$	ass cd
$x \neq i$	2 s 28 LL
$x \neq k$	19 29 LL
$\sim R(bx)$	3 s ui 30 mp cd

32. $\sim R(bx) \leftrightarrow x = j$ 21 27 cb dd

33. 32 ud

34. $\exists z \forall x [\sim R(bx) \leftrightarrow x = z]$ 20 eg cd

35. Show $k = j \rightarrow \exists z \forall x [\sim R(bx) \leftrightarrow x = z]$

show that if $k = j$ then

' $\sim R(bx)$ ' is proper

LINES 36-51 FOLLOW THE PATTERN OF 19-34

52. $k = i \vee k = j$ 2 us

53. $\exists z \forall x [\sim R(bx) \leftrightarrow x = z]$ 18 35 52 sc

54. $R(b \neg \exists x R(bx))$ 4 prd

55. $\sim R(b \neg \exists x R(bx))$ 53 prd

56. $\neg \exists x R(bx) \neq \neg \exists x \sim R(bx)$ 54 55 LL dd

conclude ' $\sim R(bx)$ ' is proper

$\leftarrow$ prd

$\leftarrow$ prd

6 INVALIDITIES WITH DEFINITE DESCRIPTIONS

Give counter-examples to show that these arguments are invalid:

1. $A \wedge \iota x A x$
 $\therefore \forall x \forall y (A x \wedge A y \rightarrow x = y)$
 Universe: $\{0, 1\}$
 A : $\{0, 1\}$
 chosen object: 0

Since 'A' is true of two things, 'Ax' is improper, and so ' $\iota x A x$ ' stands for the chosen object, 0. Since 'A' is true of 0, the premise is true. The conclusion is false because 0 and 1 are both A.

2. $\forall x \exists y R(x y)$
 $\exists x \forall y [R(x y) \rightarrow y \neq a]$
 $\therefore \iota x R(x x) = \iota x x \neq x$
 Universe: $\{0, 1, 2\}$
 R : $\{ \langle 0, 1 \rangle, \langle 1, 1 \rangle, \langle 2, 0 \rangle \}$
 a : 1
 chosen object: 0

The description ' $R(x x)$ ' is proper, since 'R' holds of exactly one pair of identical things: $\langle 1, 1 \rangle$. The definite description, ' $\iota x R(x x)$ ', then stands for 1, which is not the chosen object, 0, so the conclusion is false. The first premise is clearly true, and the second is true when 'x' is taken to be 2.

3. $\iota x A x \neq \iota x B x$
 $\iota x B x \neq \iota x C x$
 $d(a) = \iota x B x$
 $\forall x d(d(x)) = d(x)$
 $\therefore \iota x A x \neq \iota x C x$

Universe: $\{0, 1\}$
 A : $\{1\}$
 C : $\{1\}$
 B : $\{0\}$
 a : 0
 $d(0) = 0$ $d(1) = 1$

(chosen object: 0, though this is not important since all descriptions are proper)

7 COUNTER-EXAMPLES WITH INFINITE UNIVERSES

Give counter-examples with infinite universes to show that these arguments are invalid:

1. $\forall x \forall y [x \neq y \rightarrow d(x) \neq d(y)]$
 $\therefore \neg \exists x \neg \exists y x = d(y) = 1 \times x \neq x$
 Universe: $\{-1, 0, 1, 2, \dots\}$
 $d(\textcircled{1})$: $\textcircled{1}+1$
 chosen object: 0

The description ' $\neg \exists y x = d(y)$ ' in this universe is uniquely true of -1, so ' $\neg \exists y x = d(y)$ ' refers to -1, which is distinct from 0 (the chosen object), so the conclusion is false. The premise is true since whenever x and y are different, so are $x+1$ and $y+1$.

2. $\forall x R(x, c(x))$
 $\forall x \forall y \forall z [R(x, y) \wedge R(y, z) \rightarrow R(x, z)]$
 $\therefore \forall x \exists y x = 1z R(y, z)$
 Universe: $\{0, 1, 2, \dots\}$
 $R(\textcircled{1}, \textcircled{2})$: $\textcircled{1} < \textcircled{2}$
 $c(\textcircled{1})$: $\textcircled{1}+1$
 chosen object: 0

The first premise is true because everything is less than it plus one. The second premise is true because $<$ is transitive. For the conclusion, note that no matter what y is, ' $1z R(y, z)$ ' is improper, because there are things that y is less than. So ' $1z R(y, z)$ ' always refers to the chosen object, 0, and not everything is identical to 0.

3. $\forall x \forall y [b(x) = b(y) \rightarrow x = y]$
 $\exists y \forall x \neg y = 1z [z = b(x)]$
 $\therefore \exists x \exists y [x \neq y \wedge \forall z [b(z) \neq x \wedge b(z) \neq y]]$
 Universe: $\{0, 1, 2, \dots\}$
 $b(\textcircled{1})$: $\textcircled{1}+1$

The first premise, with $b(x) = x+1$, is clearly true. Notice that for any value of ' x ', the description ' $1z [z = b(x)]$ ' is proper, because there is always a unique thing got from x by adding 1 to it. And the description never stands for 0, since there is nothing in the universe that makes 0 when you add 1 to it. So the second premise is true, taking ' y ' to be 0. The conclusion is false since there aren't two things in the universe that cannot be gotten by adding 1 to something.